

Semester: IV

Course Code: 22UCSL403

Course Title: Object Oriented Programming Laboratory

Division: A (A1, A2, A3)

Academic Year: 2024-25

General instructions

- The programs must strictly follow the Java naming and coding conventions
- All the programs must use **object-oriented concepts** like classes, objects, methods, constructors, overloading, overriding, late binding, interfaces, etc. (wherever possible)
- Never write the business (core) logic inside `main()` method

Following steps are to be followed in Eclipse IDE to pass command line arguments:

- Right click on the java file from the **Package Explorer**
- Goto **Run As → Run Configurations**
- Ensure that the configuration for the selected file has been created. If not, create a run configuration by clicking on **New launch configuration** button
- Goto **Arguments** tab and enter the command line arguments separated by whitespaces inside **Program arguments** text area
- Click on **Apply** button to save the command line arguments
- Then click on **Run** button to run the program with command line arguments

Following is the procedure to read input from keyboard in Java program:

- Instantiate Scanner class:
Scanner sc = new Scanner(System.in);
- The Scanner class is available in **java.util** package. Hence import that package in the application:
import java.util.Scanner;
- To read inputs from keyboard, use one of **next()** methods. For e.g., to read an integer, following code can be used:
int n = sc.nextInt();

- Similarly, to read single precision data, use **nextFloat()** method; to read double precision data, use **nextDouble()** method.

- However, to read character data, use the following code:

```
char ch = sc.next().charAt(0);
```

- To read a string, use the following code:

```
String s = sc.next();
```

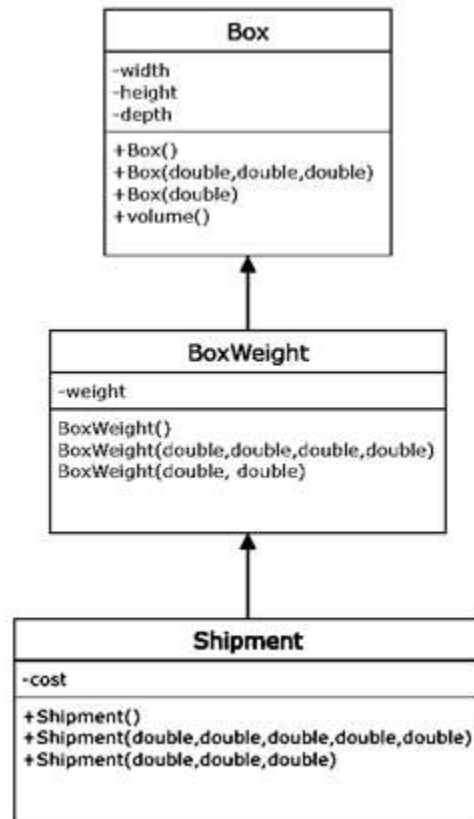
Practice Programs-1

- 1) Write a Java program that implements Simple Calculator.
- 2) Write a Java program that sorts the given array elements in ascending order (use any sorting technique of your choice). Use the following array elements:
{98.87, 45.24, 39.71, 43.89, 10.09, 47.92, 47.84, 21.17, 38.51, 62.38, 71.39, 40.34, 28.54, 72.45, 59.31, 72.39, 87.34, 58.35, 71.21}
- 3) Write a Java program that implements a stack.
- 4) Write a Java program that implements a queue.
- 5) Write a Java program that takes IA-1, IA-2, IA-3, CTA and SEE marks of n-students in a course, and displays the CIE marks and the grade obtained. Use the following table to compute the grade:

Marks (out of 100)	(90 – 100)	(80 – 89)	(70 – 79)	(60 – 69)	(50 – 59)	(45 – 49)	(40 – 44)	(39 – 0)
Grade	O	A+	A	B+	B	C	P	F

- 6) A person is recognized by first name, last name and mobile number. An employee of an organization is recognized by employee ID, designation and salary. A student of a college is recognized by USN, department and division. **Employee is-a person, Student is-a person.** Using this information, write a Java program that displays all the information of person, employee and student. Also draw the class diagram for the above scenario.
- 7) A vehicle is identified by how many wheels it has. A car is identified by engine number, chassis number and manufacturer. A bike is identified by engine number, chassis number and manufacturer. Similarly, a truck is identified by engine number, chassis number and manufacturer. **Car is-a vehicle, bike is-a vehicle and Truck is-a vehicle.** Using this information, write a Java Program that displays all the information of vehicle, car, bike and truck. Also draw the class diagram for the above scenario.

- 8) Write a Java program that implements the following class diagram and displays all information of Box, BoxWeight and Shipment classes:



- 9) Using the concept of abstract class, write a Java program that calculates the area and perimeters of circle, triangle, square and rectangle.
- 10) Using the concept of packages, implement the following scenario:

Create a package named “**basicMath**” inside “**com.sdmcet.cse.oop**” package. It has a class named **Basic** and provides following methods:

- i) Check if a number is prime or not
- ii) Finds sum of digits of a number

Create another package called “**advancedMath**” inside “**com.sdmcet.cse.oop**” package. It has a class **Advanced** and provides following methods:

- i) Calculates sine, cosine & tan of a degree
- ii) Find the sum of primary and secondary diagonal elements of a 3X3 matrix

Create a class named **MathTestDemo** defined inside “**com.sdmcet.cse.oop**” package. This class has **main()** method and must invoke all the methods of **Basic** and **Advanced** classes.

11) You are asked to develop a simple media player application that can play different types of media files such as MP3, MP4, WAV and WMA. All the listed media players must provide the common features to play, pause, stop, rewind and forward the media being streamed.

Using the feature of interfaces, write a Java program that provides the implementation for the above problem definition.

Note: No need to actually implement the above-mentioned common features, as these features require the implementation of drivers and codecs, and is beyond the scope of current discussion. Hence, it is sufficient to put relevant display statements inside the common features.
